ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6	Carbon has 6 protons.
2	**(b)** ethane	Alkane.
3	**(d)** allotropes of carbon	Both are pure carbon.
4	**(a)** 4	Tetravalent.
5	**(c)** ethene	Alkene.
6	**(c)** graphite	Graphite has free electrons.
7	**(c)** methane	CH_4.
8	**(b)** ethyne	Alkyne.
9	**(a)** diamond	Diamond.
10	**(a)** CH_4	One C, four H.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Valency is 4 (1); It has 4 electrons in its outer shell (1).	1+1
12	Same molecular formula (1); Different structures (1).	1+1
13	Alkane $CnH_{2}n_{2}^{+}$ (1); Alkene $CnH_{2}n$ (1).	1+1
14	Compound of only carbon and hydrogen (1); CH_4 and C_2H_4 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Methane CH_4 (1); Ethane C_2H_6 (1); Propane C_3H_8 (1).	1+1+1
16	Low melting and boiling points (1); Poor conductors of electricity (1); Mostly burn in air (1).	1+1+1
17	Sodium ethanoate and water (1); $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (2).	1+2
	OR: Ethene C_2H_4 (1); Propene C_3H_6 (1); Butene C_4H_8 (1).	1+1+1
18	Solvent (1); Fuel mixed with petrol (1); In medicines and tinctures (1).	1+1+1
	OR: Soaps: sodium/potassium salts of fatty acids (1); Detergents: sodium salts of sulphonic acids (1); Detergents work in hard water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CH_4: 4 C-H single bonds (1); C_2H_4: 4 C-H and a C=C (1); H_2O: 2 O-H (1); Shared pairs: 4, 6 and 2 (1); Covalent bonds form by sharing (1).	1+1+1+1+1
	OR: $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (1); $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$ (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Esterification needs conc. H_2SO_4 (1); Ester smells fruity (1).	1+1+1+1+1
20	Single bonds vs double/triple bonds (1); General formula differs (1); Saturated: clean flame (1); Unsaturated: sooty flame (1); Addition reactions in unsaturated (1).	1+1+1+1+1
	OR: Soap has two ends (1); Hydrophobic tail attaches to grease (1); Micelle forms (1); Hard water forms scum (1); Detergents solve this (1).	1+1+1+1+1

Mistakes to avoid:

- Counting covalent bonds wrongly: an alkane C_nH_{2n+2} has $3n + 1$ bonds (include all C-H and C-C bonds).
- Confusing ethanol and ethanoic acid: ethanol (C_2H_5OH) is an alcohol, ethanoic acid (CH_3COOH) releases CO_2 with carbonates; ethanol does not.
- Naming the chain from the wrong carbon count, e.g. calling CH_3CH_2CHO 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).
- Writing an addition reaction for a saturated hydrocarbon; saturated compounds undergo substitution, unsaturated ones undergo addition.

- Saying that detergents form micelles but soaps do not, or that soap works well in hard water; soaps give scum in hard water.
- Stating that diamond and graphite conduct electricity; only graphite does (it has free electrons).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** saturated hydrocarbons	Single bonds only.
2	**(b)** combustion	Combustion.
3	**(d)** C_2H_6	Two carbons single bond.
4	**(d)** propane	C_3H_8.
5	**(c)** CO_2 and water	Complete combustion.
6	**(b)** C_2H_4	One double bond.
7	**(b)** -OH	Hydroxyl.
8	**(c)** alcohol	Spirit.
9	**(c)** C_2H_2	One triple bond.
10	**(c)** -COOH	Carboxyl.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Alcohol (-OH) (1); Carboxylic acid (-COOH) (1).	1+1
12	Reaction of acid with alcohol (1); Gives ester and water (1).	1+1
13	Hydrocarbons with only single bonds (1); Ethane (1).	1+1
14	Diamond (1); Graphite (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ethyne C_2H_2 (1); Propyne C_3H_4 (1); General formula CnH_2n-2 (1).	1+1+1
16	Vinegar (1); Food preservation (1); Making esters/dyes (1).	1+1+1
17	Insoluble precipitate in hard water with soap (1); Ca and Mg salts react (1); Reduces cleaning (1).	1+1+1
	OR: -OH alcohol (1); -CHO aldehyde (1); -COOH carboxylic acid (1).	1+1+1
18	$CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ (2); Heat and light released (1).	2+1
	OR: $CH_3{-}CH_3$ (1); $CH_2{=}CH_2$ (1); $CH{\equiv}CH$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Covalent bonding (1); Low melting/boiling points (1); Poor conductors (1); Burn in air (1); Form chains (catenation) (1).	1+1+1+1+1
	OR: Diamond: hard; cutting (1); Graphite: conducts; electrodes (1); Fullerene C_{60} (1); Allotropes: same element, different structure (1); Diamond is an insulator (1).	1+1+1+1+1
20	Methane CH_4 (1); Ethane C_2H_6, propane C_3H_8 (1); Butane C_4H_{10} (1); Pentane C_5H_{12} (1); Hexane C_6H_{14} (1).	1+1+1+1+1
	OR: Addition: unsaturated compounds add atoms (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (1); Substitution: H replaced by another atom (1); $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (1); Sunlight needed for substitution (1).	1+1+1+1+1

Mistakes to avoid:

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- Naming the chain from the wrong carbon count, e.g. calling CH_3CH_2CHO 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).
- Writing an addition reaction for a saturated hydrocarbon; saturated compounds undergo substitution, unsaturated ones undergo addition.
- Saying that detergents form micelles but soaps do not, or that soap works well in hard water; soaps give scum in hard water.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** ethanoic acid	Acetic acid.
2	**(c)** C_2H_5OH	Alcohol with 2 carbons.
3	**(b)** -CHO	Formyl.
4	**(d)** fruity smell	Sweet smell.
5	**(a)** CH_3COOH	Carboxylic acid.
6	**(a)** sodium salt of fatty acid	Soap structure.
7	**(a)** nickel	Ni.
8	**(b)** sharing of electrons	Sharing of pairs.
9	**(a)** C_4H_{10}	Alkane $CnH_2n_2^+$.
10	**(c)** hard water	No scum.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Used for washing (1); Removes dirt and oil (1).	1+1
12	Contain double or triple bonds (1); Ethene (1).	1+1
13	Carbon atoms linking to form chains (1); Rings also form (1).	1+1
14	Carbon dioxide (1); Lime water turns milky (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Methanol (1); Ethanol (1); Propanol (1).	1+1+1
16	$2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$ (2); Hydrogen gas evolved (1).	2+1
17	It is the hardest known substance (1); Each carbon bonded to four others (1); Rigid 3-D structure (1).	1+1+1
	OR: Methanoic acid (1); Ethanoic acid (1); Propanoic acid (1).	1+1+1
18	$CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (2); Nickel (1).	2+1
	OR: Conducts electricity (free electrons) (1); Layers slide (1); Soft and slippery (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	n-butane (1); isobutane (2-methylpropane) (1); Structures drawn properly (1); Both are C_4H_{10} (1); Pentane has three isomers (1).	1+1+1+1+1
	OR: Formula C_2H_5OH vs CH_3COOH (1); -OH vs -COOH (1); Ethanol neutral; acid turns blue litmus red (1); Only acid gives CO_2 with carbonate (1); Uses: solvent vs vinegar (1).	1+1+1+1+1
20	With Na: sodium ethoxide and H_2 (1); $2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$ (1); With $KMnO_4$: ethanoic acid (1); With conc. H_2SO_4 at 443 K: ethene (1); Each reaction named (1).	1+1+1+1+1
	OR: Halo -Cl: CH_3Cl (1); Alcohol -OH: C_2H_5OH (1); Aldehyde -CHO: CH_3CHO (1); Ketone C=O: CH_3COCH_3 (1); Acid -COOH: CH_3COOH (1).	1+1+1+1+1

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